Elliott Salpekar

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EDUCATION

Georgia Institute of Technology **Atlanta, GA**
Bachelor of Science in Mathematics, Bachelor of Science in Computer Science *May 2027 GPA: 4.0*
Coursework: Combinatorics, Proofs, Electrostatics, OOP

Stony Brook University **Stony Brook, NY**
Bachelor of Arts in Economics*, Presidential Scholarship* *May 2025*
Coursework: Differential Equations, Multivariable Calculus, Intermediate Microeconomics
Activities and Societies: Investment Club, Quant Club, Math Club, Run Club

INDEPENDENT RESEARCH AND PROJECTS

Research Interests: Analytic modeling, Stochastic Processes, Information Theory, Physical Simulation, Time Series

Qualitative AIXI Approximator

- Theorized a practical proxy for the theoretical ideal and uncomputable reasoner called AIXI (Hutter 2005)
- Assembled a network of LLM-powered agents and a constitution-based environmental feedback system to mimic AIXI
- Formalized the information flow as uncompressed, qualitative yet high-fidelity text data for LLM comprehension

PepsiCo/Coca-Cola Cross Model and Custom Backtesting Engine

- Created a custom backtesting engine using 40+ years of historical data
- Trained and optimized a neural network using PyTorch that predicts Coca-Cola equity from PepsiCo prices
- Integrated predictions into a trading strategy with consistent Sharpe Ratios of 1.2+

Custom Retail Analytics Suite

- Created a series of analytics tools to identify customer behaviors, price items, and inform stocking decisions
- Used forecasting techniques such as Factorization Machines, Transformers, and Kaplan-Meier Survival Analysis
- Automated dataflow from a popular POS system's API and integrated features into a user-friendly HTML frontend

PROFESSIONAL

OxBlue **Atlanta, Georgia**
Technology Intern *December 2024 – January 2025*

- Independently ideated and Programmed an automated procedure and Excel workspace to prepare billing data, reducing manual effort by 70%
- Cold-called 100+ clients to collect overdue invoices
- Shadowed CFO to learn corporate finance workflows

PERSONAL

World Travel *August 2023 – August 2024*

- Circumnavigated the world on personal savings over the course of a year
- Bicycled independently for 1400+ miles across Cambodia and Vietnam
- Volunteered in diverse disciplines such as childcare, farming, and education
- Summited El Chachani, Peru (20,000 ft above sea level)

LEADERSHIP AND SERVICE

Piano Performance and Recording Club *August 2024 – Present*
Founder and President

- Initiated a student organization which provides recording and performance resources to the pianist community
- Secured $1000+ in institutional funding for professional recording equipment

Azul Wasi Orphanage *June 2023 – July 2024*

- Taught Spanish to native Quechua-speaking orphaned students and prepared large group meals

Kad Sala English Camp *February 2024 – March 2024*

- Prepared and led regular English classes with children and adults and produced online educational content

SKILLS

Computer: Deep learning, Cloud Computing, Python, PyTorch, HTML, Microsoft Office, Virtual Machines, LaTeX
Language: Proficient in Spanish, Native Proficiency in English
Special: Distance running including regular marathons, Piano performance and composition, Nutritional sciences

AWARDS

- 11th place nationwide out of 2500+ contestants in Skew The Script's 2023 High School Data Science Challenge
- First prize in Music Composition section of the Ithaca High School newspaper 'Literary Issue' two consecutive years.